

Rank-Before-Merge isolates the ranking stage

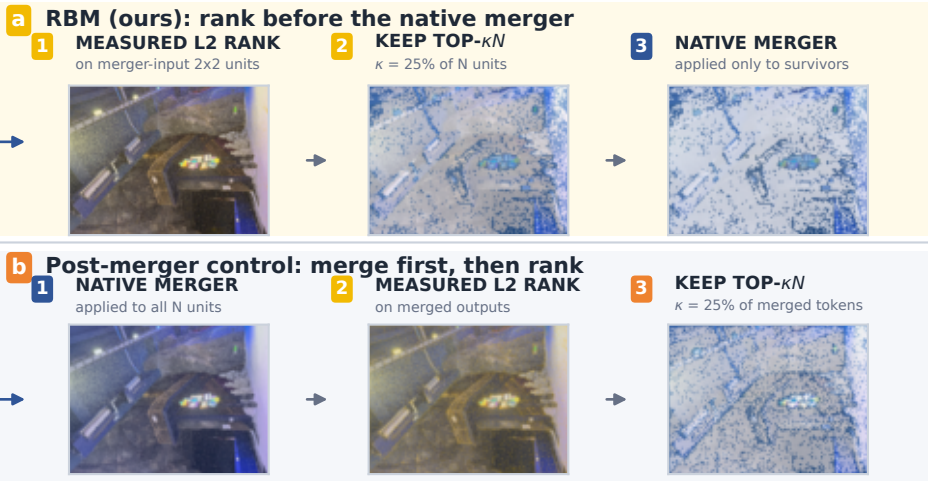
Same input, budget, L2 selector, and native merger

Shared visual input



One image, two controlled paths

ViT units (N)
merger input



■ solid border: kept

□ faded: discarded

■ blocky tile: merged token

Controlled comparison: same image, model path, κ , L2 selector, and native merger; only the score tap point changes.

Qwen3-VL-8B | measured merger features | 25% kept

color: per-arm min-max | selection: raw measured L2 (NPZ keep)

No attention or query signal is depicted.